PROJECT DEVELOPMENT & MANAGEMENT PLAN

Wildfire Risk Prediction Using Climate & Environmental Data

Master of Data Science | Final Year Project | 26 May 2026 – 15 July 2026

Student Name	[Your Full Name]
Programme	Master of Data Science
Project Title	Wildfire Risk Prediction Using Climate & Environmental Data
Dataset	1.88M US Wildfires — Kaggle (rtatman/188-million-us-wildfires)
Models	Ridge Classifier, MLP Neural Network, SVM (RBF Kernel), Gaussian Naive Bayes
Tuning Method	Hyperband Optimization
Imbalance	Balanced Bagging Classifier
Explainability	ELI5 (Explain Like I'm 5)
Evaluation	Accuracy, F1, ROC-AUC, PR Curve, 5-Fold Stratified CV
Start Date	26 May 2026
Deadline	15 July 2026 — 7 Weeks Total
Supervisor	[Supervisor Name]

1. Research Questions

Main Research Question:

"To what extent can machine learning models accurately predict wildfire risk classification using historical climate and environmental data?"

Sub-Questions:

- RQ1: Which model — Ridge Classifier, MLP Neural Network, SVM (RBF Kernel), or Gaussian Naive Bayes — achieves the highest ROC-AUC score for wildfire risk prediction?
- RQ2: Which environmental features are the most significant predictors of high-risk wildfire events, as identified through ELI5 explainability analysis?
- RQ3: How does Balanced Bagging Classifier improve F1-score and precision-recall performance on the imbalanced wildfire dataset?
- RQ4: To what degree does Hyperband hyperparameter tuning improve MLP Neural Network performance compared to its default configuration?

2. Project Objectives

- Download the 1.88M US Wildfire dataset directly from Kaggle using the kagglehub API.
- Clean and preprocess the data — handle missing values, encode categories, remove duplicates.
- Engineer new features: SEASON, RISK_BINARY, FIRE_SIZE_LOG, and REGION.
- Apply Balanced Bagging Classifier to handle class imbalance in the training data.
- Perform EDA with 5 charts and 1 interactive geospatial Folium heatmap.

- Train and compare 4 classification models: Ridge Classifier (baseline), MLP Neural Network, SVM (RBF Kernel), Gaussian Naive Bayes.
- Fine-tune MLP Neural Network using Hyperband (30 trials) to maximise ROC-AUC score.
- Evaluate all models using Accuracy, F1, ROC-AUC, Precision-Recall Curve, and 5-Fold Stratified Cross-Validation.
- Apply ELI5 to explain model predictions and identify top environmental predictors of wildfire risk.
- Complete and submit the full academic thesis by 15 July 2026.

3. Detailed Week-by-Week Plan

Week	Dates	Phase	Tasks	Deliverable
1	26 May – 1 Jun	**Project Setup**	Set up Google Colab, configure Kaggle API credentials (os.environ), install all libraries (kagglehub, keras-tuner, eli5, folium, imbalanced-learn, scikit-learn). Download dataset using kagglehub. Auto-detect SQLite table name. Inspect all available columns.	**Environment ready + dataset downloaded**
2	2 Jun – 8 Jun	**Data Cleaning**	Handle missing values in all columns. Drop rows with missing FIRE_SIZE_CLASS, LATITUDE, LONGITUDE, STATE, STAT_CAUSE_DESCR. Fill OWNER_DESCR nulls with UNKNOWN. Remove duplicates. Reset index. Validate — 0 missing values in key columns.	**Clean dataset ready**
3	9 Jun – 15 Jun	**Feature Engineering + EDA**	Create SEASON from day-of-year, RISK_BINARY (0=Low, 1=High), FIRE_SIZE_LOG (log transform), REGION (geographic grouping). Apply Balanced Bagging Classifier to handle class imbalance. Scale features with StandardScaler. Produce 5 EDA plots + Folium geospatial heatmap of High Risk zones.	**Feature matrix + 6 EDA plots saved**
4	16 Jun – 22 Jun	**Model Training**	Train Model 1: Ridge Classifier (baseline). Train Model 2: MLP Neural Network (3 hidden layers, ReLU activation). Train Model 3: SVM with RBF Kernel (trained on 100K sample due to size). Train Model 4: Gaussian Naive Bayes. Record Accuracy, F1, ROC-AUC for all 4 models.	**4 trained models + scores**
5	23 Jun – 29 Jun	**Hyperband Tuning + Evaluation**	Fine-tune MLP Neural Network using Hyperband (30 trials) — tune units per layer, learning rate, dropout rate, number of layers. Run 5-fold stratified cross-validation on all models. Plot confusion matrix, ROC curve, precision-recall curve. Apply ELI5 to explain predictions. Print all classification reports.	**Tuned MLP + full evaluation plots**
6	30 Jun – 6 Jul	**Thesis Writing Ch. 1–3**	Save all models using joblib and keras save(). Write Chapter 1 (Introduction — wildfire problem, motivation, research gap). Write Chapter 2 (Literature Review — cite all 10 references). Write Chapter 3 (Dataset & EDA — explain all visualisations and data findings).	**Ch. 1–3 drafted**
7	7 Jul – 15 Jul	**Thesis Completion + Submission**	Write Chapter 4 (Methodology — justify all 9 techniques). Write Chapter 5 (Results — compare all 4 models, Hyperband tuning results). Write Chapter 6 (Discussion — limitations, climate impact, feature importance analysis). Write Chapter 7 (Conclusion — contributions and future work). Proofread full document. Submit to supervisor by 15 July 2026.	**FINAL SUBMISSION ✓**

4. Milestone Tracker

Date	Milestone	Description	Status

1 Jun 2026	Dataset Ready	Downloaded via kagglehub, table detected, columns validated	■ Pending
8 Jun 2026	Data Clean	All missing values handled, duplicates removed	■ Pending
15 Jun 2026	EDA Complete	All 5 charts + Folium heatmap saved, Balanced Bagging applied	■ Pending
22 Jun 2026	4 Models Trained	Ridge, MLP, SVM, Gaussian Naive Bayes trained and scored	■ Pending
29 Jun 2026	Tuning + Eval Done	Hyperband complete, all evaluation plots and ELI5 analysis done	■ Pending
6 Jul 2026	Ch. 1–3 Done	Introduction, Literature Review, Dataset chapters written	■ Pending
15 Jul 2026	FINAL SUBMISSION	All 7 chapters complete and submitted to supervisor	■ Pending

5. Techniques We Are Using & Why

No.	Technique	Why We Are Using This
1	**Ridge Classifier**	Used as our baseline model. It is a linear classifier that applies L2 regularisation to prevent overfitting. Fast to train and easy to compare against — all other models must beat it to prove their value.
2	**MLP Neural Network**	Multi-Layer Perceptron captures complex non-linear patterns between climate features like temperature, season, and location. It is our most powerful model and is fine-tuned with Hyperband to find the optimal architecture.
3	**SVM (RBF Kernel)**	Support Vector Machine with RBF kernel finds the optimal decision boundary between Low Risk and High Risk fires in high-dimensional feature space. Academically strong and widely cited in classification research.
4	**Gaussian Naive Bayes**	A probabilistic classifier that predicts risk based on the probability of each feature given the class. Fast, interpretable, and provides a very different approach from the other three models for strong comparison.
5	**Balanced Bagging Classifier**	Wraps our base models and automatically balances the 90/10 class imbalance by undersampling the majority class in each bootstrap sample. Cleaner and more robust than oversampling techniques like SMOTE.
6	**Hyperband**	A modern hyperparameter tuning algorithm that stops poor-performing trials early and allocates more resources to promising ones. Much faster than Grid Search and smarter than random search for tuning the MLP Neural Network.
7	**ELI5**	ELI5 — Explain Like I'm 5 — explains model predictions in plain English. It shows which features pushed a prediction towards High Risk or Low Risk, making our model transparent and easy to present to non-technical stakeholders.
8	**5-Fold Stratified CV**	Divides the dataset into 5 equal folds and tests the model 5 times on different splits. Stratified ensures each fold has the same High/Low Risk ratio. Confirms our results are stable and not a lucky outcome from one split.
9	**Folium Heatmap**	Creates an interactive geographic map showing High Risk fire zones across the United States. Red areas indicate highest danger density. Makes results accessible to fire agencies and policy makers beyond technical audiences.

6. Risk Plan

Risk	Level	Mitigation

Kaggle API token fails or expires	**Medium**	Regenerate token from Kaggle Settings page and re-run the credential cell
SVM too slow on 1.88M records	**High**	Train SVM on a 100K stratified sample — still valid for comparison
MLP Neural Network overfits on training data	**Medium**	Add Dropout layers (0.3) and Early Stopping callback during training
Hyperband tuning takes too long on Colab	**Medium**	Reduce max_epochs to 20 and use Colab GPU runtime (T4)
Colab session disconnects during training	**High**	Save each model immediately with joblib or keras model.save() after training
ELI5 not compatible with all sklearn models	**Low**	Use eli5.explain_weights() for linear models, eli5.explain_prediction() for others
Thesis writing falls behind schedule	**High**	Write one chapter per day in Week 6 and Week 7 — stick to the plan

7. Sign-Off

Student Name	
Student Signature	________________________________ Date: ___________
Supervisor Name	
Supervisor Signature	________________________________ Date: ___________
Submission Deadline	15 July 2026
Status	■ Approved ■ Revise & Resubmit

8. References

[1] [Dataset] Short, K. C. (2017). Spatial wildfire occurrence data for the United States, 1992–2015 (FPA FOD 20170508). Forest Service Research Data Archive. https://doi.org/10.2737/RDS-2013-0009.4

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[6] [Ridge Classifier] Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. Technometrics, 12(1), 55–67. https://doi.org/10.1080/00401706.1970.10488634

[7] [Gaussian Naive Bayes] Zhang, H. (2004). The optimality of Naive Bayes. Proceedings of the 17th International FLAIRS Conference, 562–567. AAAI Press.

[8] [Wildfire ML Review] Jain, P., Coogan, S. C., Subramanian, S. G., Crowley, M., Taylor, S., & Flannigan, M. D. (2020). A review of machine learning applications in wildfire science and management. Environmental Reviews, 28(4), 478–505. https://doi.org/10.1139/er-2020-0019

[9] [Climate & Wildfire] Abatzoglou, J. T., & Williams, A. P. (2016). Impact of anthropogenic climate change on wildfire across western US forests. Proceedings of the National Academy of Sciences, 113(42), 11770–11775. https://doi.org/10.1073/pnas.1607171113

[10] [Wildfire Prediction ML] Sayad, Y. O., Mousannif, H., & Al Moatassime, H. (2019). Predictive modeling of wildfires: A new dataset and machine learning approach. Fire Safety Journal, 104, 130–146. https://doi.org/10.1016/j.firesaf.2019.01.006

PDM prepared for supervisor presentation — Master of Data Science Final Year Project